

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

Gravitation

Q1

A satellite of mass 'M' is projected radially from surface of earth with speed 'u'. When it reaches a height equal to radius of earth, it ejects a rocket of mass $\frac{M}{10}$ and it itself starts orbiting the earth in circular path of radius $2R$, find the kinetic energy of rocket.

एक द्रव्यमान 'M' के उपग्रह को पृथ्वी की सतह से 'u' चाल के साथ त्रिज्या दिशा में प्रक्षेपित किया जाता है। जब वह पृथ्वी की त्रिज्या के बराबर ऊँचाई तक पहुँच जाता है, तब रॉकेट $\frac{M}{10}$ द्रव्यमान को निष्कासित कर देता है और यह स्वयं $2R$ त्रिज्या के वृत्तीय पथ में परिक्रमा करना शुरू कर देता है, तो रॉकेट की गतिज ऊर्जा होगी

- (1) $5M\left(u^2 - \frac{119GM_e}{200R}\right)$ (2) $5M\left(u^2 - \frac{113GM_e}{200R}\right)$ (3) $\frac{M}{20}\left(u^2 - \frac{119GM_e}{100R}\right)$ (4) $\frac{M}{20}\left(u^2 - \frac{113GM_e}{200R}\right)$

7th Jan Morning

Sol

(1)

$$\frac{-GM_e M}{R} + \frac{1}{2}Mu^2 = \frac{-GM_e M}{2R} + \frac{1}{2}Mv^2$$



$$v = \sqrt{u^2 - \frac{GM_e}{R}}$$

Q2

If weight of an object at pole is 196 N then weight at equator is [$g = 10 \text{ m/s}^2$; radius of earth = 6400 Km]
यदि एक वस्तु का भार ध्रुव पर 196N है, तो भूमध्य रेखा पर इसका भार होगा। [$g = 10 \text{ m/s}^2$; पृथ्वी की त्रिज्या = 6400 Km]

- (1) 194.32 N (2) 194.66 N (3) 195.32 N (4) 195.66 N

7th Jan Evening

Sol

(3)

at pole, weight ध्रुव पर भार = $mg = 196$

$$m = 19.6 \text{ kg}$$

at equator, weight भूमध्य रेखा पर भार = $mg - m\omega^2 R$

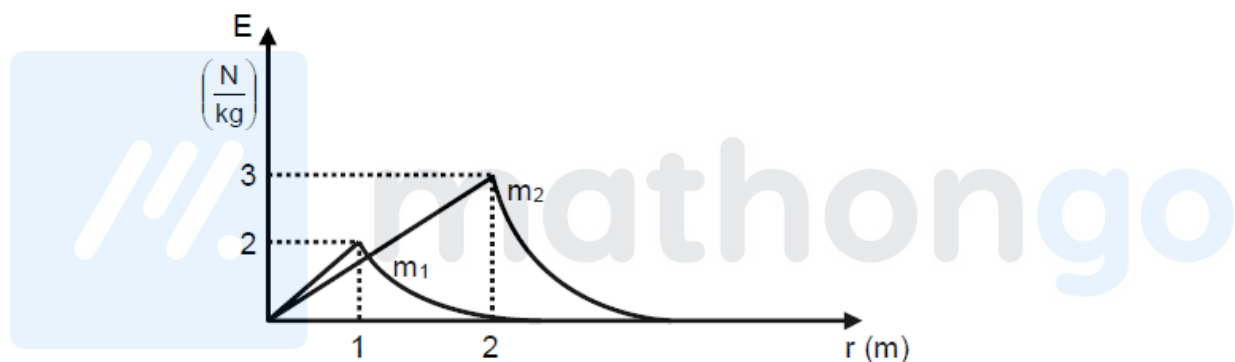
$$= 196 - (19.6) \left[\frac{2\pi}{24 \times 3600} \right]^2 \times 6400 \times 10^3$$

$$= 195.33 \text{ N}$$

Q3

Two spherical bodies of mass m_1 and m_2 have radii 1 m and 2 m respectively. The gravitational field of the two bodies with the radial distance from centre is shown below. The value of $\frac{m_1}{m_2}$ is—

दो गोलाकार वस्तुओं का द्रव्यमान क्रमशः m_1 एवं m_2 तथा इनकी त्रिज्या क्रमशः 1 m व 2 m है। इन वस्तुओं के लिए केन्द्र से त्रिज्यीय दूरी के साथ गुरुत्व क्षेत्र की तीव्रता का वक्र चित्र में प्रदर्शित है। $\frac{m_1}{m_2}$ का मान ज्ञात करो।



(1) $\frac{1}{6}$

(2) $\frac{1}{3}$

(3) $\frac{1}{2}$

(4) $\frac{1}{4}$

8th Jan Morning

Sol

(1)

$$3 = \frac{Gm_2}{2^2}$$

$$2 = \frac{Gm_1}{1^2}$$

$$\therefore \frac{3}{2} = \frac{1}{4} \frac{m_2}{m_1}$$

$$\frac{m_1}{m_2} = \frac{1}{6}$$

Q4

A particle is dropped from height $h = 100$ m, from surface of a planet. If in last $\frac{1}{2}$ sec of its journey it covers 19 m. Then value of acceleration due to gravity that planet is :

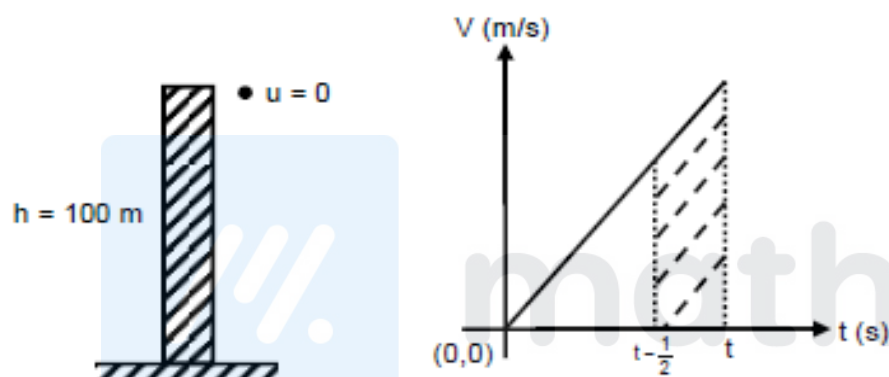
किसी ग्रह के सतह से $h = 100$ m की ऊँचाई से एक कण छोड़ा गया है। अपनी यात्रा के अन्तिम $\frac{1}{2}$ सेकण्ड में 19 m दूरी तय करता है। इस ग्रह पर गुरुत्वीय त्वरण का मान होगा।

- (1) 8 m/s^2 (2) $\frac{1}{8} \text{ m/s}^2$ (3) $\frac{1}{4} \text{ m/s}^2$ (4) 2 m/s^2

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Sol

(1)



Area of shaded trapezium

छायांकित भाग का क्षेत्रफल

$$= \frac{g \left[t - \frac{1}{2} + t \right]}{2} \times \frac{1}{2} = 19 \quad \dots(1)$$

$$\frac{1}{2}gt^2 = 100 \quad \dots(2)$$

$$\Rightarrow t = \sqrt{\frac{200}{g}}$$

$$g \left[2t - \frac{1}{2} \right] = 76 \quad \Rightarrow \frac{76}{g} = \frac{\left[4\sqrt{\frac{200}{g}} - 1 \right]}{2}$$

$$g = 8 \text{ m/s}^2$$

Q4

Gravitation**JEE Mains 2020 Jan Chapter wise Question Bank**

An asteroid of mass m ($m \ll m_E$) is approaching with a velocity 12 km/s when it is at distance of $10 R$ from the centre of earth (where R is radius of earth). When it reaches at the surface of Earth, its velocity is (Nearest Integer) in km/s .

8th Jan Evening**Sol****16**

$$KE_i + PE_i = KE_f + PE_f$$

$$\frac{1}{2} m u_0^2 + \left(-\frac{GMm}{10R} \right) = \frac{1}{2} m v^2 + \left(-\frac{GMm}{R} \right)$$

$$v^2 = u_0^2 + \frac{2GM}{R} \left[1 - \frac{1}{10} \right]$$

$$v = \sqrt{u_0^2 + \frac{9}{5} \frac{GM}{R}}$$

$$= \sqrt{12^2 + \left(\frac{9}{5} \right) \frac{(11.2)^2}{2}}$$

$$= \sqrt{144 + 0.9(11.2)^2} = \sqrt{256.896}$$

$$= 16.028 \text{ km/s}$$

$$\simeq 16$$

Q5

A particle of mass m is revolving around a planet in a circular orbit of radius R . At the instant the particle has velocity $\vec{V}$, another particle of mass $\frac{m}{2}$ moving at velocity $\frac{\vec{V}}{2}$ collides perfectly in-elastically with the first particle. The new path of the combined body will take is

- | | |
|-------------------|---------------------------------------|
| (1) Circular | (2) Elliptical |
| (3) Straight line | (4) Fall directly below on the ground |

9th Jan Morning**Sol**

(2)

Conserving momentum संवेग संरक्षण से:

$$\frac{m}{2} \frac{v}{2} + mv = \left(m + \frac{m}{2}\right) v_f$$

$$v_f = \frac{5mV}{4 \times \frac{3m}{2}} = \frac{5V}{6}$$

 $v_f < v_{orb} (= v)$ thus the combined mass will go on to an elliptical path $v_f < v_{orb} (= v)$ अतः संयुक्त द्रव्यमान दीर्घवृत्ताकार पथ पर गति करेगा।

Q6

Two planets of masses M and $\frac{M}{2}$ have radii R and $\frac{R}{2}$ respectively. If ratio of escape velocities from their surfaces $\frac{v_1}{v_2}$ is $\frac{n}{4}$, then find n :

दो ग्रह जिनके द्रव्यमान M तथा $\frac{M}{2}$ तथा त्रिज्यायें क्रमशः R व $\frac{R}{2}$ है। यदि इनके लिये सतह से पलायन वेगों का अनुपात

$\frac{v_1}{v_2}$ का मान $\frac{n}{4}$ है तो n ज्ञात करे -

(1) 3

(2) 1

(3) 2

(4) 4

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Sol

(4)

$$v_e = \sqrt{\frac{2GM}{R}}$$

$$\therefore \frac{v_1}{v_2} = \frac{\sqrt{\frac{2GM}{R}}}{\sqrt{\frac{2GM/2}{R/2}}} = 1 = \frac{n}{4}$$

$$\Rightarrow n = 4$$

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